Sound Zones - Spectral Analysis Project

Stationary and Non-stationary Spectral Analysis

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1 Introduction

Imagine having a room where multiple speakers are placed. The speakers are all playing the same audio, let's say it's a piece of music you want to enjoy. Studying one point P in the room, the sound from each speaker will have traveled different distances before arriving at that point. This means the audio arrives with different phases from different speakers and also with different amplitudes since the amount of sound absorption in the air depends on the distance the sound has traveled. Thus the sound arriving from the different speakers might cancel each other out in some points P, while they instead amplify each other at other points. In this project, we wish to create a filter for each speaker that phase shifts and amplifies the different frequency components of the original audio signal in such a way that constructive interference happens in one predefined zone of the room and destructive interference happens in another. The zone where we wish constructive interference to happen is called the bright zone and here, one wishes to hear the original audio. The zone of destructive interference is called the dark zone and, optimally, this zone would be silent.

1.1 The Room Impulse Response

Since a sound played by a speaker will not only travel the direct and closest way to a point P, but also bounce on any walls or other objects in the room a number of times, a way of efficiently describing how the sound actually behaves from a speaker s to a microphone m is needed. The room impulse response does exactly this. As the name suggests, it means an impulse is played by a speaker s and a microphone m is recording what it hears. This recording is called the room impulse response and often denoted as $h_{m,s}(t)$.

The room impulse response is a powerful tool. For example, it can be shown that convoluting an audio $x_s(t)$ with the room impulse response $h_{m,s}(t)$ gives you the audio $y_m(t) = h_{s,m}(t) * x_s(t)$ which microphone m would hear if speaker s would play $x_s(t)$. Note that there is one room impulse response $h_{m,}(t)$ per microphone-speaker pair (m, s). Applying a Fourier Transform to the room impulse response, we get a complex-valued vector $H_{m,s}(f)$ describing the change in amplitude and phase shift happening between speaker s and microphone m per frequency. Since a convolution in the time domain is a multiplication in the frequency domain, we can write $Y_m(f) = X_s(f) \cdot H_{m,s}(f)$. Studying only one frequency f, the complex number $H_{m,s}(f)$ tells us the change in amplitude $|H_{m,s}(f)|$ and phase $\angle H_{m,s}(f)$ that tone will experience when played by speaker s and recorded by microphone m.

Now, let's increase from only studying one speaker-microphone pair to a total of S number of speakers in a room. In the time domain, the audio heard by microphone m when each speaker plays an audio $x_s(t)$ can be described by a sum of the convolution of that audio with each room impulse response. We write

$$y_m(t) = \sum_{s=1}^{S} h_{m,s}(t) * x_s(t).$$

In the frequency domain, one would write the equivalent expression as

$$Y_m(f) = \sum_{s=1}^{S} H_{m,s}(f) \cdot X_s(f)$$

for each frequency.

Finally, we increase to also studying a total number of M microphones. Instead of studying an audio per speaker $x_s(t)$, let's assume all speakers are playing the same audio $x(t)$, but with some change of amplitude and phase to each frequency component. The change of amplitude and phase that speaker s makes to the original audio at frequency f is saved in the complex number g_s and can be viewed as a filter of that speaker at that certain frequency. All filters are saved in a vector $\mathbf{g}(f)$. Multiplying $\mathbf{g}(f)$ with the room impulse responses $H_{m,s}(f)$ gives us

$$\mathbf{p}(f) = \mathbf{H}(f) \cdot \mathbf{g}(f) = \begin{bmatrix} p_1(f) \\ \vdots \\ p_M(f) \end{bmatrix} = \begin{bmatrix} H_{1,1}(f) & \cdots & H_{1,S}(f) \\ \vdots & \ddots & \vdots \\ H_{M,1}(f) & \cdots & H_{M,S}(f) \end{bmatrix} \begin{bmatrix} g_1(f) \\ \vdots \\ g_S(f) \end{bmatrix}$$

where $p_m(f)$ is a complex number telling us the change in amplitude $|p_m(f)|$ and phase $\angle p_m(f)$ that microphone m hears when the filter $\mathbf{g}(f)$ is applied to all speakers.

Finally, doing this for every frequency f_i in a frequency grid of n number of frequencies, one gets

$$\mathbf{p_m} = \begin{bmatrix} p_m(f_1) \\ p_m(f_2) \\ \vdots \\ p_m(f_n) \end{bmatrix},$$

which is the sound that microphone m hears, described in the frequency domain, meaning it's the Fourier transform of the audio $y_m(t)$. Therefore, applying an inverse Fourier Transform to $\mathbf{p_m}$ would give us $y_m(t)$, the sound actually heard by microphone m when all the filters $\mathbf{g}$ have been applied to the speakers.

1.2 Finding Optimal Speaker-filters

There are two main algorithms for achieving sound zones: Acoustic Contrast Control (ACC) and Pressure Matching (PM). In this investigation, ACC was initially explored, but ultimately PM was used.

The PM method aims to perfectly reproduce a desired sound field in the bright zone at full strength, while maintaining silence in the dark zone. To do this, we divide our previously defined room impulse response matrix $\mathbf{H}$ into two smaller matrices: $\mathbf{H}_\mathrm{b}$ for the microphones located in the bright zone, and $\mathbf{H}_\mathrm{d}$ for the microphones in the dark zone. We also define a target sound field $\mathbf{p}_\mathrm{des}$, which describes the ideal amplitude and phase we wish to achieve at each microphone in the bright zone.

The goal is to find the optimal speaker filters $\mathbf{g}$ that minimize the difference between the reproduced sound field in the bright zone $\mathbf{H}_\mathrm{b}\mathbf{g}$ and the desired sound field $\mathbf{p}_\mathrm{des}$. However, this must be done while keeping the sound energy in the dark zone below a certain threshold D_0, and ensuring the energy exerted by the speakers does not exceed a limit E_0, as otherwise the algorithm might find unrealistic solutions of infinite energy.

This PM formulation can be written mathematically as an optimization problem:

$$\begin{aligned} \min_{\mathbf{g}} \quad & \|\mathbf{H}_\mathrm{b}\mathbf{g} - \mathbf{p}_\mathrm{des}\|^2 \\ \text{subject to} \quad & \|\mathbf{H}_\mathrm{d}\mathbf{g}\|^2 \leq D_0 \\ & \|\mathbf{g}\|^2 \leq E_0 \end{aligned}$$

To solve this constrained optimization problem, it is rewritten as a Lagrangian cost function. By introducing the Lagrange multipliers λ_1 and λ_2, we can combine the objective and the constraints into a single equation:

$$\min_{\mathbf{g}} L_p(\mathbf{g}) = \|\mathbf{H}_\mathrm{b}\mathbf{g} - \mathbf{p}_\mathrm{des}\|^2 + \lambda_1(\|\mathbf{H}_\mathrm{d}\mathbf{g}\|^2 - D_0) + \lambda_2(\|\mathbf{g}\|^2 - E_0), \quad \lambda_1, \lambda_2 \geq 0$$

The variables λ_1 and λ_2 act as tuning weights. λ_1 determines how heavily we penalize sound leaking into the dark zone, while λ_2 penalizes excessive speaker effort to prevent hardware distortion or unreasonable filter weights.

The solution that minimizes this cost function L_p is obtained by setting the derivative of L_p with respect to $\mathbf{g}$ to zero, which results in the following linear system:

$$[\mathbf{H}_\mathrm{b}^H\mathbf{H}_\mathrm{b} + \lambda_1\mathbf{H}_\mathrm{d}^H\mathbf{H}_\mathrm{d} + \lambda_2\mathbf{I}]\mathbf{g} = \mathbf{H}_\mathrm{b}^H\mathbf{p}_\mathrm{des}$$

Here, $\mathbf{I}$ is the identity matrix, and $\mathbf{H}^H$ denotes the Hermitian transpose of a matrix. This equation has to be solved for each frequency to obtain the optimal filter weights $\mathbf{g}$ to apply to the speakers. In this investigation, we set the regularization parameters to $\lambda_1 = 1$ (applying equal effort to matching the pressure in the bright zone and minimizing the energy in the dark zone) and $\lambda_2 = 0.01$.

To achieve a coherent and natural-sounding audio reproduction in the bright zone, the desired sound field $\mathbf{p}_\mathrm{des}$ was modeled as a plane wave propagating through the bright zone. A direction of propagation is needed for a plane wave and the angle $\theta = 0$ was set to correspond to a wave moving from left to right in the room. The desired pressure at the m-th bright zone microphone, located at coordinates (x_m, y_m), was defined as

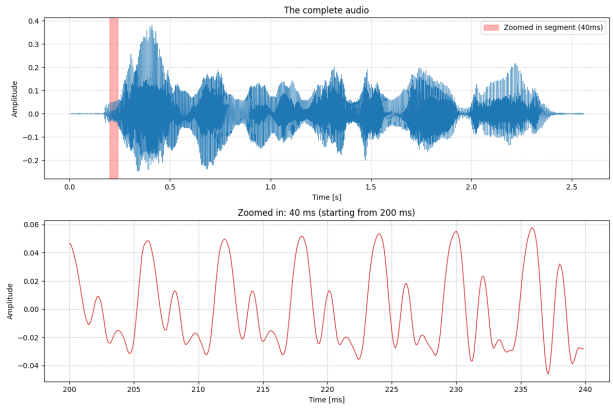
$$p_\mathrm{des}^m = e^{-j(k_x x_m + k_y y_m)}$$

in the frequency domain, where $k_x = k\cos(\theta)$ and $k_y = k\sin(\theta)$ are the directional components of the wave vector, and $k = 2\pi f/c$ is the wavenumber for frequency f. By formulating $\mathbf{p}_\mathrm{des}$ as a propagating plane wave, the algorithm aims to recreate a phase-coherent wavefront.

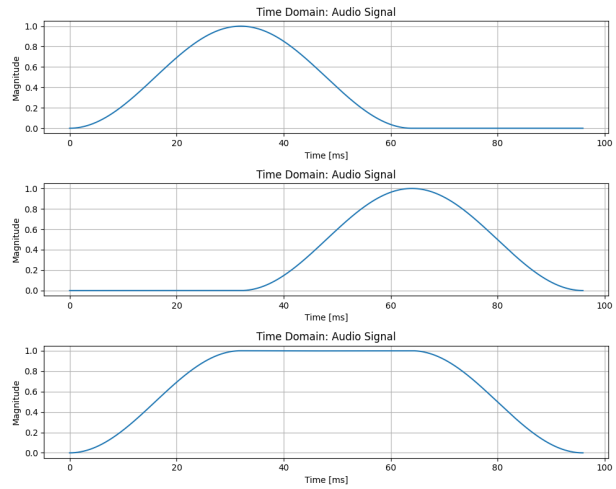
1.3 Splitting the Audio and Stitching it Together

When processing speech, one usually looks at one window at the time of between 20 and 40 ms. Namely, a chunk of that length of human speech can be said to be stationary, see for example figure 1a. Splitting the

full audio into chunks like this makes it possible to analyze the spectral components of each syllable. Since the spectral content will change over time, the previously mentioned filters applied to the speakers should also change with time. Then, one filter can be applied to one chunk, another filter to the next chunk and so on. To make sure the transition between the applied filters are smooth, an overlap of chunks is desired with some added window. A Hanning window with 50% overlap was applied, since this enables smooth transitions while all windows still sum to one, see figure 1b.



(a) The complete audio of a woman saying "Why were you away a year Roy?" and a zoomed in syllable of 40 ms.

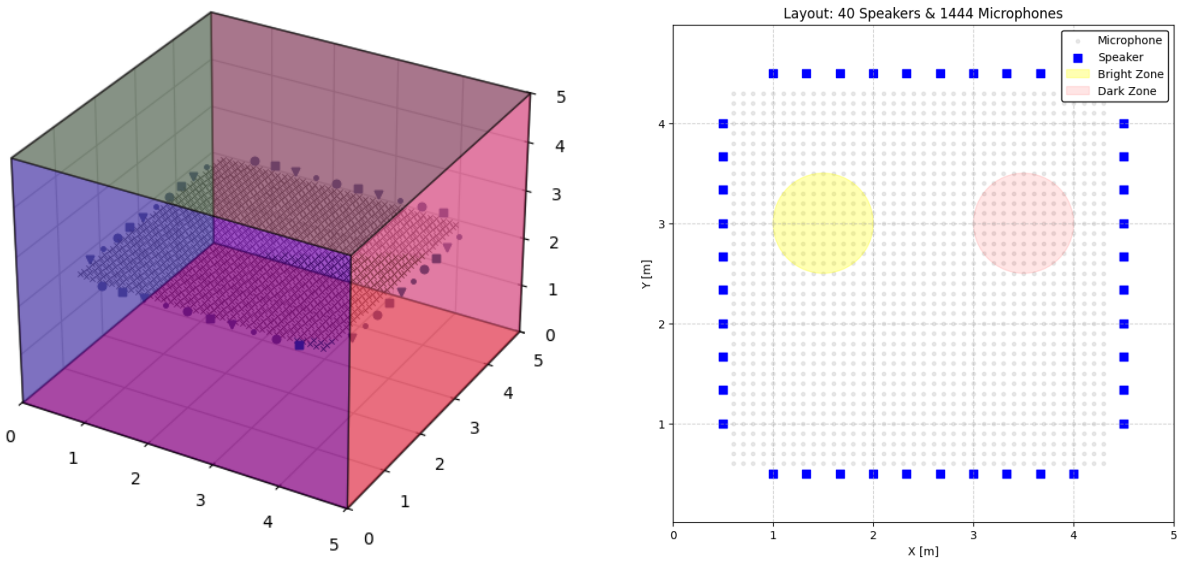


(b) Two Hanning-windows with 50% overlap. The plot shows they add to one.

Figure 1

2 Method and Results

The chosen dimensions for the room in which to simulate the sound zones was a cube with side lengths of five meters. The python package *pyroomacoustics* gave the options of an Anechoic room or a ShoeBox. In the Anechoic room, the walls completely absorb all sound and therefore there is no echo. In the Shoebox room the echo is taken into account, and the number of bounces for each audio was set to 3. Air absorption is present in both types of rooms, which means the intensity of a sound will decrease with the distance it has traveled. The setup of the speakers, microphones, and sound zones in the room was set as seen in Figure 2. To simplify working with a room in three dimensions, all speakers and microphones were set in the same z-plane. Ten speakers were set along each of the four side walls and a grid of microphones with the spacing of 0.1 m was placed between them. The Bright Zone center was placed at $(1.5, 3.0)$ and the Dark Zone center at $(3.5, 3.0)$, both having a radius of 0.5 m. The radius was later decreased to 0.1 m.



(a) The used room geometry. All microphones and speakers are set in one plane, $z = 2.5$ m.

(b) The speaker and microphone setup in 2D, where $z = 2.5$ m. The bright and dark zones are here set as circles with 1 m in diameter.

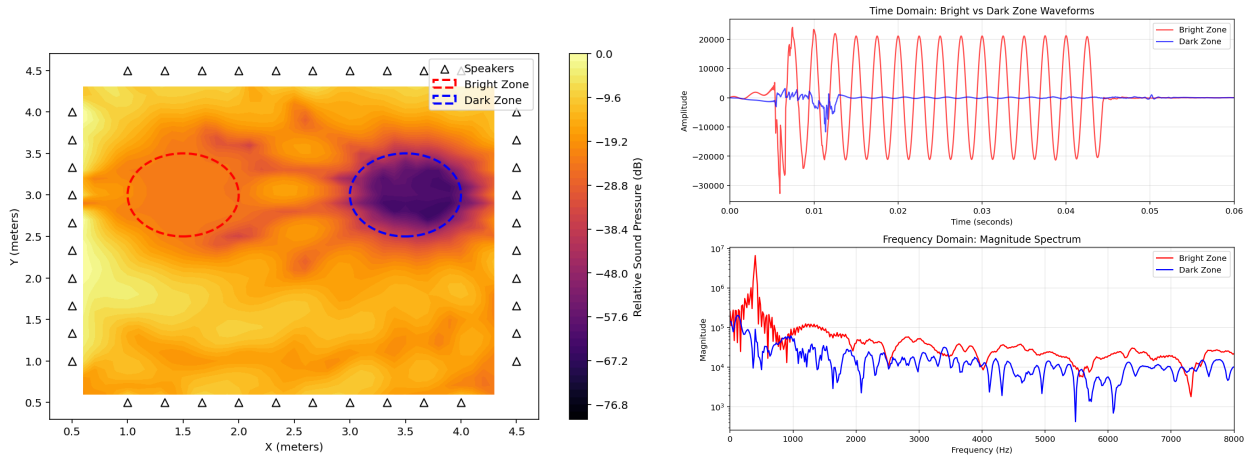
Figure 2: Room Setup

2.1 Single Frequency in Anechoic Room

With the aim of studying the simplest possible problem first, we started by creating an Anechoic room and generated a single-frequency signal at 400 Hz. Since there is no reflection of the sound on any of the walls in an anechoic room, the frequency domain of the room impulse response $H(m, s, f)$ can be calculated easily from the distance $r_{m,s}$ between the speaker s and the microphone m since both the amplitude decay and the phase shift will depend on this distance. The calculation is done using the free-field Green's function

$$H(m, s, f) = \frac{1}{r_{m,s}} e^{-j\frac{2\pi f}{c} r_{m,s}}$$

Having calculated the room impulse response, the Pressure Matching algorithm was used to calculate the filter vectors $\mathbf{g}(f)$. The resulting spatial sound pressure level is shown in Figure 3. This setup achieved an acoustic contrast of 36 dB between the center of the Bright Zone and the center of the Dark Zone.



(a) Acoustic Pressure Map of Anechoic room (dB)

(b) Time-domain and frequency-domain of simluated signal over 60 ms.

Figure 3: Acoustic pressure map (dB) and simulated signal in an anechoic room using Pressure Matching.

To visualize the resulting spatial sound field, the total relative sound pressure level (SPL) is calculated. The total energy E at each microphone position is calculated as

$$E_m = |A(f) \cdot p_m(f)|^2$$

where $A(f)$ represents the complex amplitude of the original audio signal of frequency f and $p_m(f)$ is the complex pressure change at microphone m at that same frequency. The Root-Mean-Square (RMS) pressure is then converted to a logarithmic scale (dB) and normalized to the maximum observed value in the grid:

$$L_{dB,m} = 20 \log_{10} \left(\sqrt{E_m} + \epsilon \right) - \max(L_{dB})$$

where ϵ is a small constant (10^{-12}) to prevent numerical instability. This normalization allows for easier evaluation of the acoustic contrast between the zones independent of the source amplitudes.

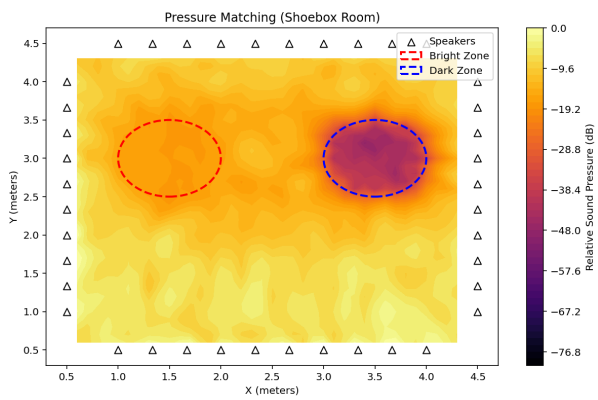
2.2 Multiple Frequencies in a Shoebox Room

Following the successful proof-of-concept in an Anechoic environment, the model was extended to a Shoebox room. The introduction of reflective walls requires the use of the full Room Impulse Response to construct the transfer matrix $\mathbf{H}$. As a step towards real-world signals, such as human speech, a simulation with multiple frequencies was conducted using the frequency ratios of major chord, in this case 400, 500, and 600 Hz. This approach requires calculating a unique control vector $\mathbf{g}(f)$ for each of the three frequency components.

To visualize the resulting spatial sound field, the total relative sound pressure level (SPL) is now calculated as the sum of the pressure magnitudes of each frequency. For a set of frequencies Ω, the total energy E at each microphone position is calculated as

$$E_m = \sum_{f \in \Omega} |A(f) \cdot p_m(f)|^2$$

The sound field can be seen in Figure 5a and resulted in an acoustic contrast of 2.6 dB between the zone centers. By simulating the propagation using the *pyroomacoustics* package, the signals captured by the microphones at the centers of the Bright and Dark zones were extracted, as shown in Figure 4b. In the frequency domain, there are peaks in the 400 - 600 Hz range corresponding to the major chord, hinting that the pressure matching algorithm successfully reconstructed the intended spectral content within the Bright zone while suppressing it, although much more moderately now, in the Dark zone. This was also corroborated by listening to the signals, where the major chord could still be clearly heard.



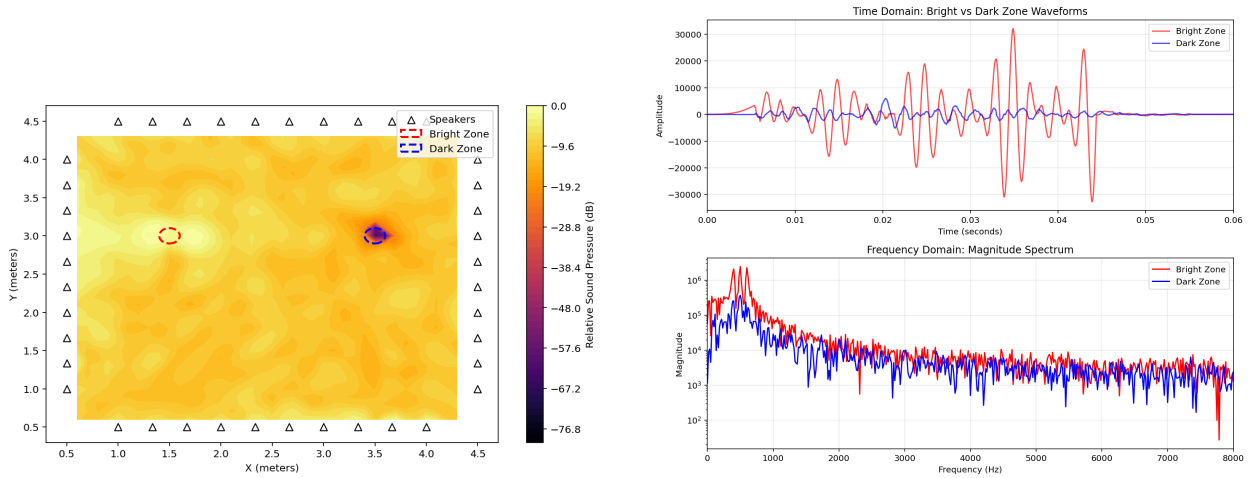
(a) Relative Sound Pressure Level (SPL) of the major chord signal.

(b) Time and frequency domain signals at the zone centers.

Figure 4: Acoustic analysis of a three-frequency major chord (400, 500, 600 Hz) in a Shoebox room.

2.3 Multiple Frequencies in a Shoebox Room - Small Zones

Exactly the same setup was used as in the previous section, with the only difference being the size of the zones. This was done with the purpose of investigating how the zone size affects the ability of the algorithm to achieve acoustic contrast. Now, both the bright and the dark zones have a radius of 0.1 m instead of 0.5 m, as it was before. The acoustic contrast between the centers of the two zones are now 14.6 dB.



(a) Relative Sound Pressure Level (SPL) of the major chord signal.

(b) Time and frequency domain signals at the zone centers.

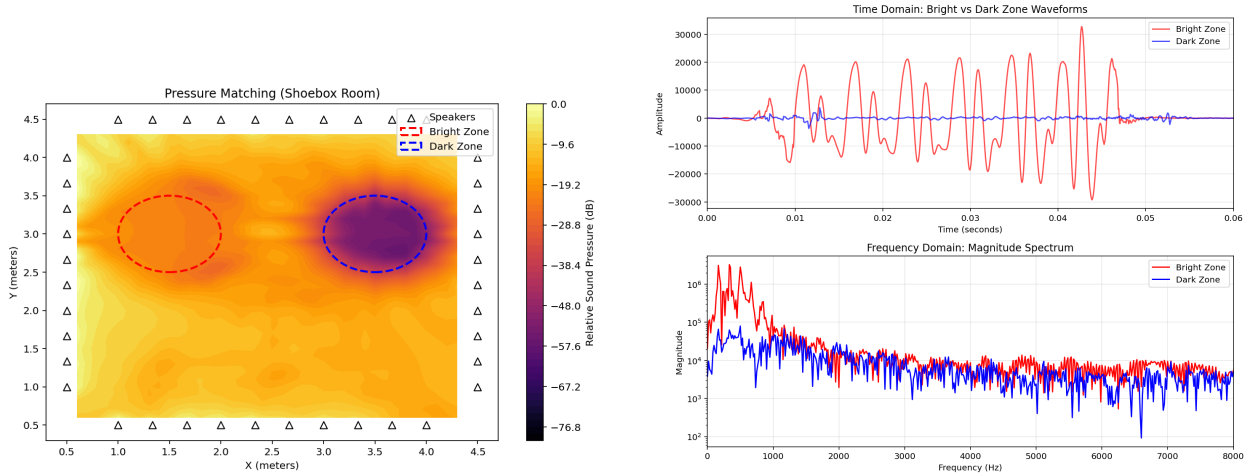
Figure 5: Acoustic analysis of a three-frequency major chord (400, 500, 600 Hz) in a Shoebox room.

2.4 Analysis of a 40 ms Speech Segment in Anechoic Room

Transitioning to real speech introduces new complexities, primarily because speech is a non-stationary signal. To address this, the audio is analyzed in short 40 ms frames, a duration over which speech can be reasonably approximated as wide-sense stationary. Furthermore, unlike the discrete frequencies of the major chord, the spectral content of speech is complex and continuously distributed.

Consequently, the signal is transformed into the frequency domain using a FFT, evaluating frequencies up to the Nyquist limit, in this case 8 kHz. The frequency resolution is determined by the sampling rate divided by N_{FFT}, with $N_{FFT} = 1024$. Just as in the multi-frequency case, the total spatial sound field and the broadband acoustic contrast are calculated by summing the energy contributions across all frequency bins.

For this simulation, the well-known audio file "Why were you away a year, Roy?" was utilized. An initial 40 ms segment, extracted from $t = 0.20$ s to $t = 0.24$ s, was processed. The resulting spatial sound pressure level and the signals captured at the zone centers are presented in Figure 6. For this specific analysis window, the algorithm achieved a broadband acoustic contrast of approximately 25 dB. It should be noted, however, that this will fluctuate significantly depending on the audio chunk being processed.



(a) Relative Sound Pressure Level (SPL) of the 40 ms speech segment.

(b) Time and frequency domain signals at the zone centers.

Figure 6: Acoustic analysis of a 40 ms speech extract in a Anechoic room.

While these initial results were promising, it was noticed that the contrast wasn't equally good throughout the signal, specifically worse during the first and last 15 ms. This discrepancy arises from the physical propagation delay of sound. The PM algorithm calculates control vectors assuming an established, steady-state sound field where all speaker contributions act on the microphones simultaneously. In reality, sound takes time to travel from the perimeter speakers to the listening zones.

Given a maximum speaker-to-microphone distance of approximately 4.8 m and the speed of sound at roughly 343 m/s, the maximum propagation delay is roughly 0.014 s (14 ms). Consequently, it takes approximately 15 ms for the wavefronts from the furthest speakers to arrive and for the intended interference pattern to fully form. This transient effect is visible in Figure 6b, where the amplitude in the Dark Zone is not properly suppressed during the initial 15 ms, but flattens out once the steady-state condition is met. A similar breakdown occurs at the end of the frame as the closest speakers cease transmitting before the furthest ones do.

To mitigate these effects, time-domain windowing was implemented using the Hanning window described in the introduction. By tapering the amplitude of the signal at the start and end of the frame, the transient errors are suppressed while preserving the steady-state performance in the center of the window. Applying this technique resulted in a visibly cleaner Dark Zone signal, as shown in Figure 7, and improved the broadband acoustic contrast to 31.8 dB.

2.5 Analysis of a 40 ms Speech Segment in a Shoebox Room

Applying the time-windowed method within Shoebox room initially yielded a significantly reduced acoustic contrast of approximately 4 dB. The introduction of wall reflections once again severely degrades the ability of the pressure matching algorithm to maintain a quiet Dark Zone over a large spatial area. While this performance drop could be mitigated by increasing the number of loudspeakers or optimizing their placement, a simpler approach was adopted for this stage of the study: reducing the radius of the control zones.

The zones were resized to a radius of 0.1 m, roughly approximating the size of a human head. With this

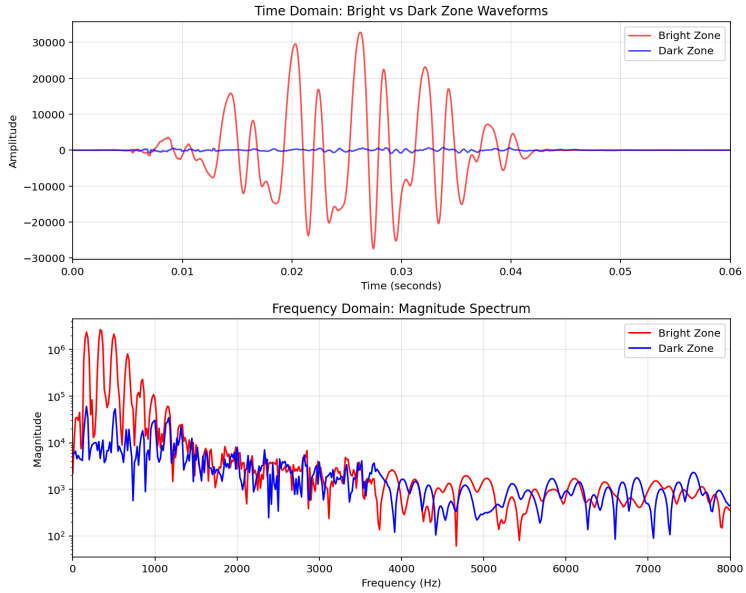
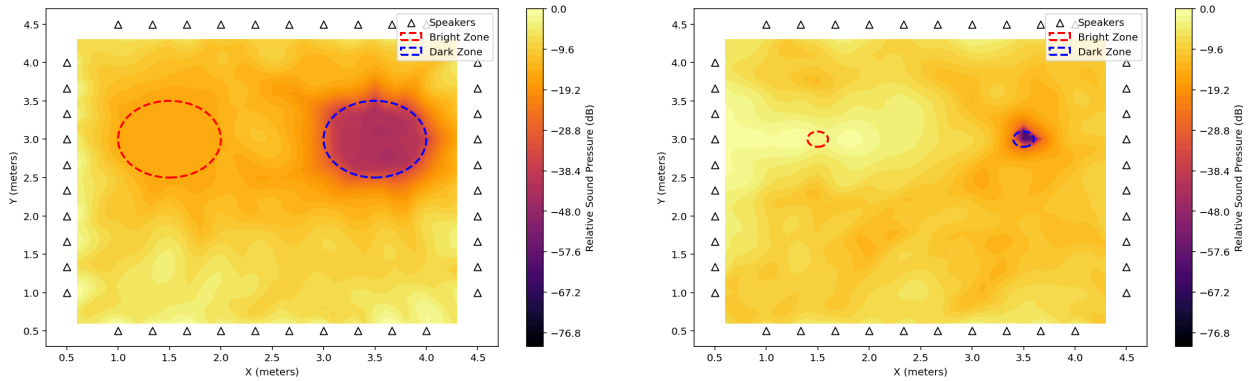


Figure 7: Time and frequency domain of windowed signals at the zone centers, showing reduced transient leakage.

modification, the broadband acoustic contrast improved to 14.4 dB. As illustrated in Figure 8, narrowing the spatial constraints allows the algorithm to focus its control effort more effectively, resulting in a clearly audible level of acoustic isolation despite the reverberant environment.



(a) Original zone radius.

(b) Reduced zone radius ($r = 0.1$ m).

Figure 8: Comparison of the resulting sound field of Speech Segment and acoustic contrast in a reverberant Shoebox room when modifying the control zone sizes.

2.6 Analysis of Full Speech Signal

In this stage, the objective was to process the entire speech signal (approximately 2 seconds) rather than an isolated 40 ms segment. This was achieved by segmenting the audio and stitching the processed chunks back together using an overlap-add methodology. The processing pipeline for a single chunk is illustrated in

Figure 9.

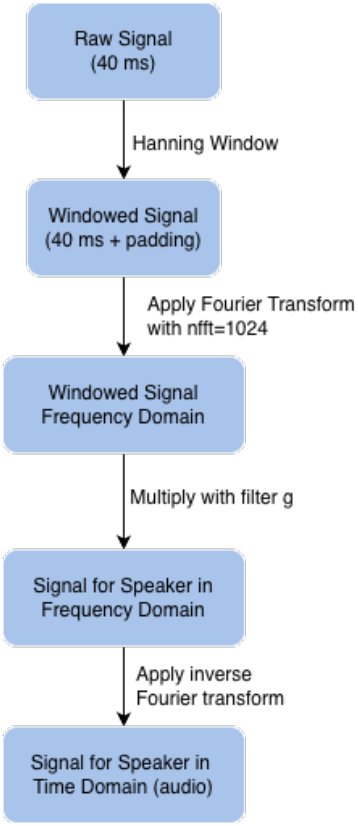
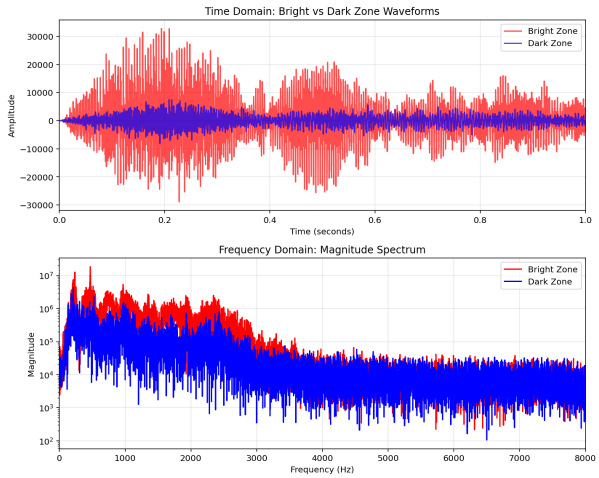


Figure 9: Explanatory diagram of the signal processing pipeline for a single audio segment.

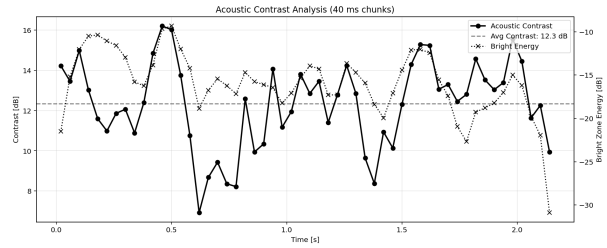
To ensure a smoother reconstruction, consecutive 40 ms chunks were extracted with a 50% overlap. As seen in section 2.3, the windowing will add up to unity in the overlapping sections. Consequently, the hop size (the start time) for each successive chunk advances by 20 ms. Furthermore, applying the spatial filters in the frequency domain extends the time-domain signal to the full FFT length ($N_{FFT} = 1024$). Retaining this full overlapping "tail" rather than strictly truncating the audio at 40 ms resulted in a better-sounding output.

The simulated signals captured at the zone centers are presented in Figure 10a. In the time domain, the Dark Zone exhibits a consistently lower amplitude throughout the duration of the audio. In the frequency domain, the Bright Zone magnitude remains distinctly higher than that of the Dark Zone, with the attenuation being particularly successful at lower frequencies (0–3 kHz) compared to higher frequencies (3–8 kHz). The broadband acoustic contrast across the entire signal was approximately 12.5 dB, producing a clearly audible difference.

However, the acoustic contrast is not constant over time. To analyze this, the contrast was evaluated over discrete 40 ms intervals, as shown in Figure 10b. The plot reveals a positive correlation between the Bright Zone signal energy and the achieved acoustic contrast. This behaves as expected: during moments of high signal energy (e.g., vowels or louder speech), the algorithm has more amplitude to create interference. Conversely, during low-energy unvoiced consonants or silences (such as the pause following the word "Roy"), both the energy and the contrast drop significantly, as there is simply less acoustic energy to cancel.



(a) Time and frequency domains of half the processed speech signal.



(b) Time-varying acoustic contrast vs. Bright Zone signal energy.

Figure 10: Comprehensive acoustic analysis of the full speech signal.

3 Further Research

There are several improvements or avenues for further research that could be explored from this basis, either to improve the listening experience or to enhance the acoustic contrast, or both. Here, we will outline spatial smoothing, the choice of $\mathbf{p}_{\text{des}}$, and audio quality.

3.1 Smoother Sound Zones

In the standard Pressure Matching algorithm, there are two primary constraints: the total energy in the dark zone and the total energy exerted by the speakers. A third constraint one might desire is spatial smoothness, ensuring that adjacent microphones do not receive vastly different sound energies. This is important because a listener moving their head slightly within the bright zone should not experience a sudden and significant change in loudness.

A possible solution, which was initially explored but not fully investigated, is encoding this "smoothness" as a third penalty term in the Lagrangian cost function, controlled by its own tuning factor, λ_{smooth}. For example, one could penalize the pressure differences between adjacent microphones using a spatial Laplacian matrix, $\mathbf{L}$. Mathematically, this introduces a new term to the linear system used to solve for the optimal speaker filters $\mathbf{g}$:

$$[\mathbf{H}_{\text{b}}^H\mathbf{H}_{\text{b}} + \lambda_1\mathbf{H}_{\text{d}}^H\mathbf{H}_{\text{d}} + \lambda_2\mathbf{I} + \lambda_{\text{smooth}}\mathbf{H}_{\text{b}}^H\mathbf{L}\mathbf{H}_{\text{b}}]\mathbf{g} = \mathbf{H}_{\text{b}}^H\mathbf{p}_{\text{des}}$$

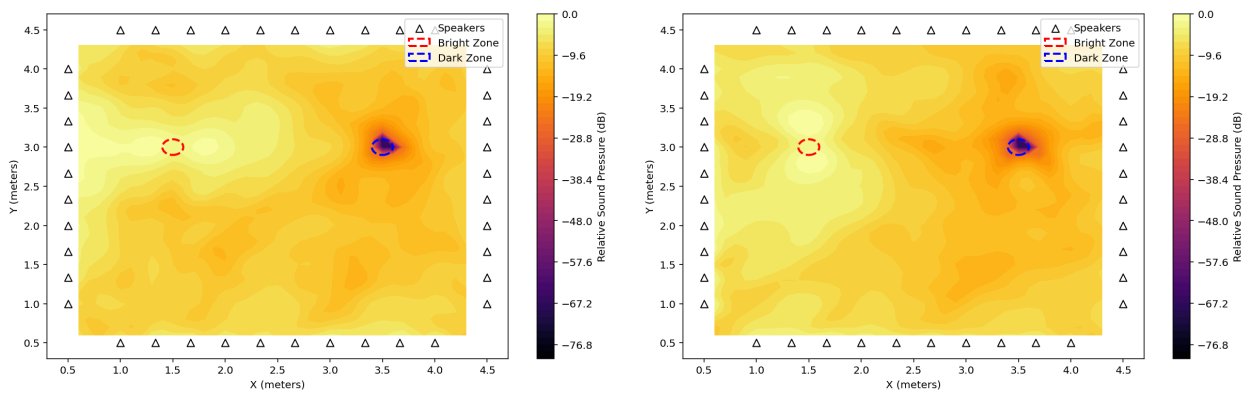
By adding the term $\lambda_{\text{smooth}}\mathbf{H}_{\text{b}}^H\mathbf{L}\mathbf{H}_{\text{b}}$ to the regularization matrix, the algorithm is forced to balance the original constraints with the new goal of avoiding sharp, unnatural spatial pressure gradients across the bright zone.

3.2 Choice of Target Sound Field, $\mathbf{p}_{\text{des}}$

In the PM algorithm, the goal is to have the bright zone accurately reproduce a desired target sound field, $\mathbf{p}_{\text{des}}$. For this investigation, the simple (and somewhat arbitrary) case of a plane wave propagating horizontally from left to right ($\theta = 0$) was chosen. However, this is by no means the optimal $\mathbf{p}_{\text{des}}$, as the ideal target field heavily depends on the geometry of the room and the specific use case for the sound zones. Even in our hypothetical setup, a very straightforward improvement is to choose a more strategic direction of propagation.

For example, taking the reverberant Shoebox scenario from Section 2.4, we can simply change the wave's direction of propagation from left-to-right ($\theta = 0$) to bottom-to-top ($\theta = \pi/2$). As shown in Figure 11, this small adjustment yielded an improvement in the broadband acoustic contrast from 12.5 dB to 14.3 dB.

The reason for this improvement becomes clear when observing the resulting spatial sound fields. In Figure 11a, the algorithm creates a bright energy "channel" along the x-axis to synthesize the left-to-right wave. Because the dark zone is positioned directly to the right of the bright zone, this horizontal channel naturally leaks more acoustic energy into the area we want to keep quiet. By rotating the target wave by 90 degrees, as seen in Figure 11b, the energy channel is oriented along the y-axis, safely bypassing the dark zone and thereby increasing the acoustic isolation.



(a) Propagation Left-Right ($\theta = 0$)

(b) Propagation Bottom-Top ($\theta = \pi/2$)

Figure 11: Comparison of the spatial sound fields based on the chosen direction of propagation for the desired plane wave, $\mathbf{p}_{\text{des}}$.

3.3 Real Speech Audio Quality

If one were to simply truncate real speech into isolated 40 ms segments and concatenate them, the resulting audio would sound highly robotic and noisy due to abrupt boundary discontinuities. Implementing a Hanning window with a 50% overlap-add (OLA) methodology significantly improved the perceived audio quality by smoothing the amplitude transitions between adjacent segments.

Despite this, noticeable sound quality degradation remains. This is especially prominent in the dark zone audio. In the PM formulation, the bright zone is constrained to a coherent target phase and amplitude ($\mathbf{p}_{\text{des}}$), whereas the dark zone lacks an explicit target sound field; the objective is merely to minimize the total energy present in the zone. Consequently, the solver is mathematically free to introduce extreme, disorganized phase shifts across frequency bins to force destructive interference. When transformed back

into the time domain, these rapid phase variations can manifest as temporal smearing and a less clean sound.

There could be ways to mitigate this degradation in future iterations. Different windowing functions could be evaluated to further reduce spectral leakage beyond the Hanning one. Also, temporal smoothing could be applied to the control filters, $\mathbf{g}$, preventing the speaker weights from jumping drastically between consecutive 40 ms frames. Finally, introducing some sort of soft phase constraint in the dark zone could lead the algorithm towards a cleaner sound.